

Management of patient with cardiovascular disease



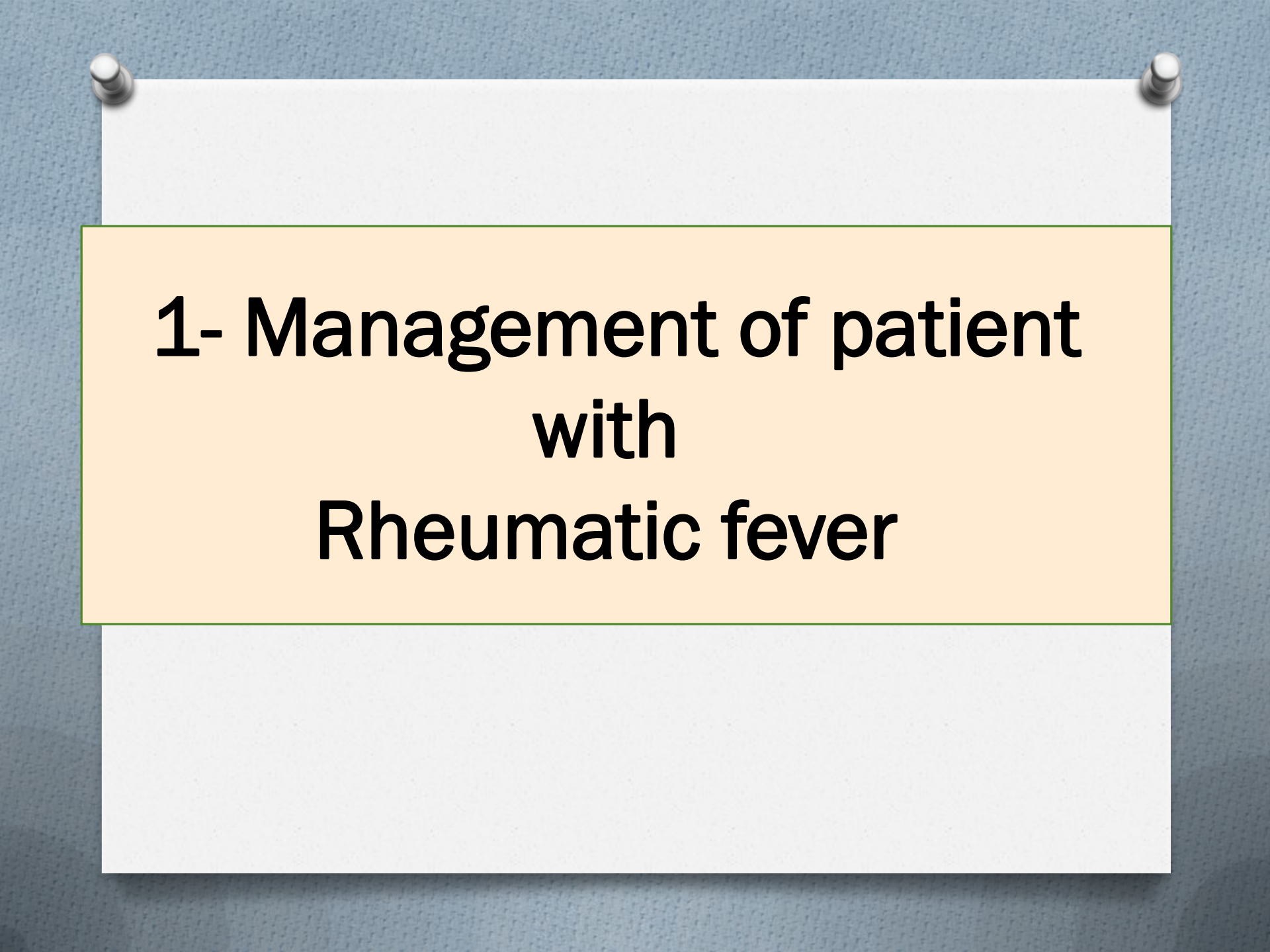
Dr . Bardes Shaban



الحديث الجانبي يشتت الانتباه



من فضلك اغلق الهاتف



1- Management of patient with Rheumatic fever

OUTLINES

- ◉ Definition
- ◉ Etiology and Pathogenesis
- ◉ Clinical Features
- ◉ Diagnosis
- ◉ Complications
- ◉ Treatment
- ◉ Prevention

◎ Definition:

Rheumatic fever is an inflammatory disease that can develop as a complication of inadequately treated strep throat which caused by an infection with group A streptococcus bacteria.

Incidence

- Rheumatic fever occurs as an **autoimmune** reaction to an **upper** respiratory (throat) streptococci infection .
- From **2:4** weeks **after** the streptococcal infection.

RISK FACTORS: -

- ⦿ Patient with recurrent infection (sore throat and tonsillitis).
- ⦿ It typically occurs between ages 5 and 15
- ⦿ Malnutrition, overcrowding
- ⦿ low socioeconomic status.

DIAGNOSIS: -

"Jones Criteria

2 major criteria

or

1 major + **2** minor criteria.

Major criteria

Carditis

Arthritis

Chorea

Subcutaneous nodules

Erythema marginatum

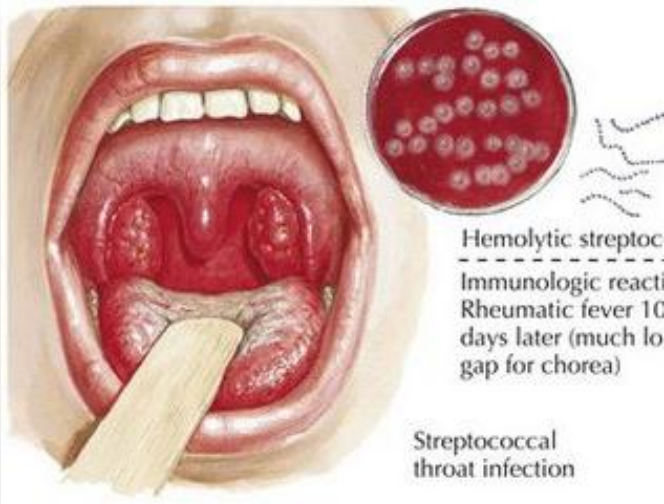
Minor criteria

Fever

Joint pain

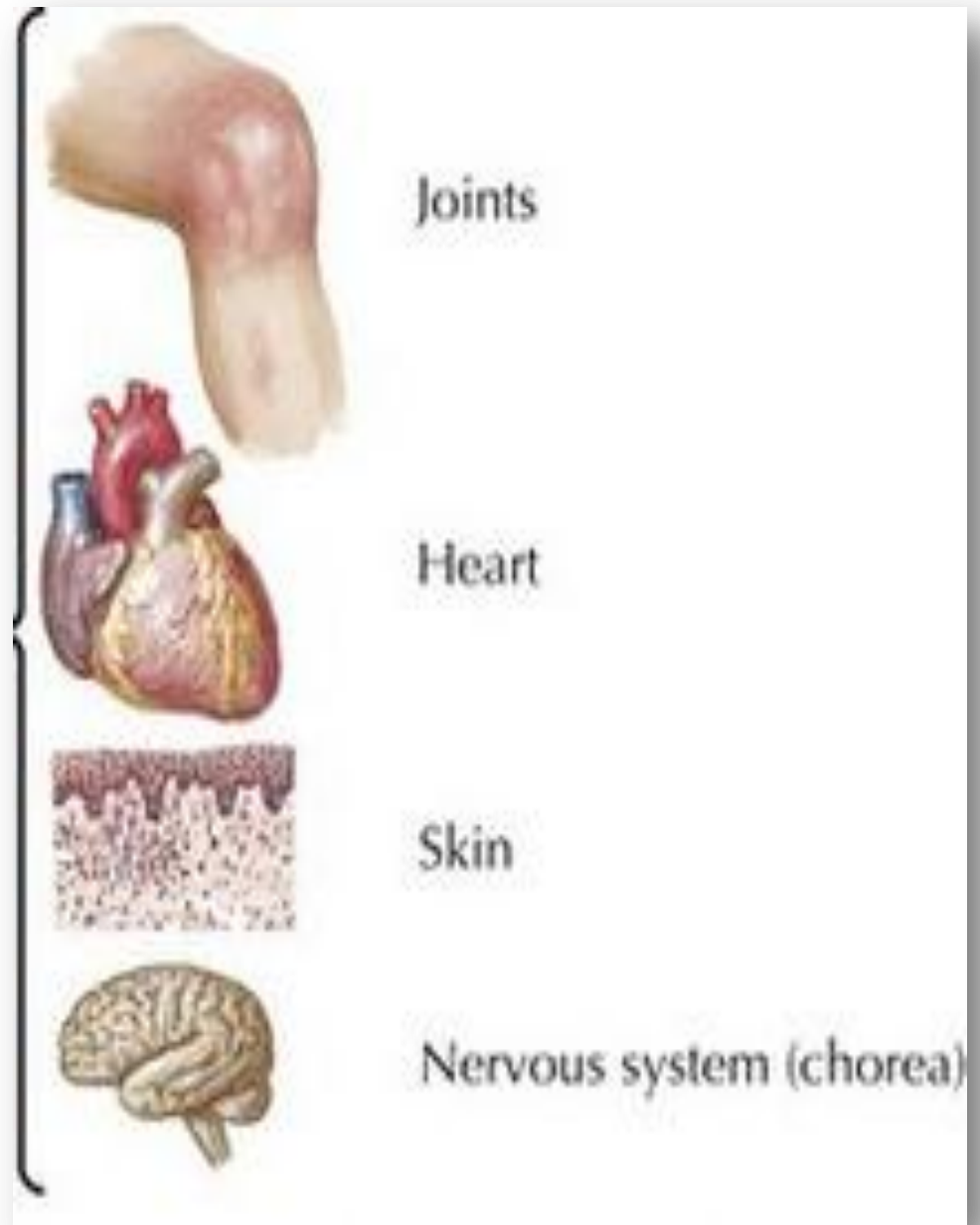
Abnormalities of ECG

Presence of certain
protein in blood



Hemolytic streptococci
Immunologic reaction
Rheumatic fever 10
days later (much longer
gap for chorea)

Streptococcal
throat infection

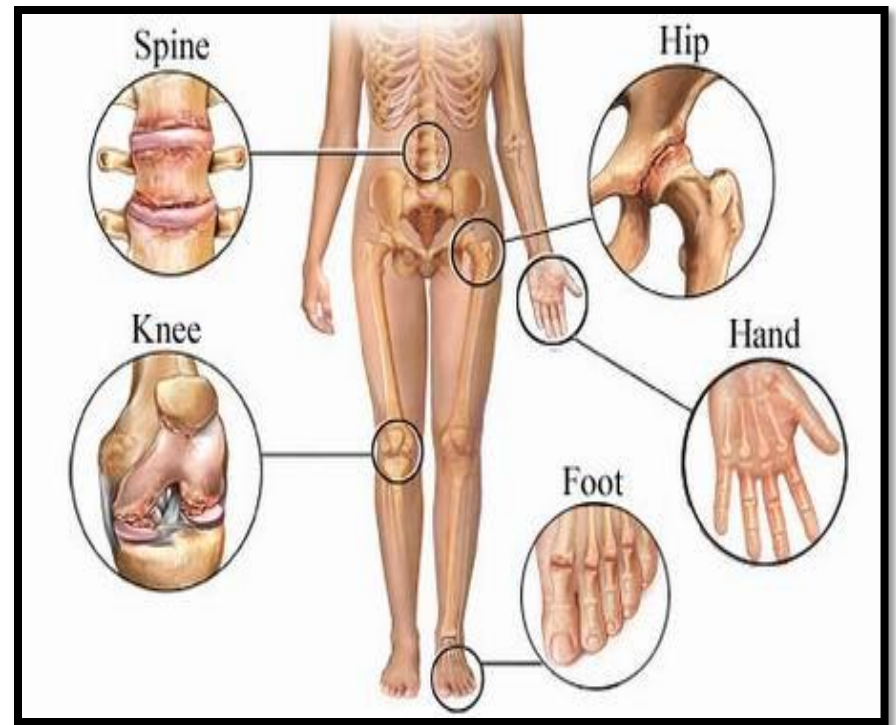


Joints

Heart

Skin

Nervous system (chorea)



Diagnostic assessment: -

- ❖ **Positive** throat **culture** for streptococci.
- ❖ **Increase** in **cardiac enzymes**.
- ❖ **Positive** C-reactive protein, **ESR** which is elevated in inflammation
- ❖ **Increase** in WBC
- ❖ **Echocardiogram** to assess for damage to valves

Treatment

- Administer non **steroidal anti-inflammatory** medication (Aspirin)
- Administer **antibiotics** (Erythromycin, penicillin)
- If damage of heart valves occur, **valve replacement** is required
- **Anticoagulation** if atrial fibrillation.

Nursing Intervention

- Monitor for difficulty breathing (**dyspnea**) ,**hacking**, **nonproductive cough**, as these are signs of heart failure.
- Determine if patient is **allergic** to penicillin.
- Monitor **S&S** of infection because RF may recur:
 - Red, sore throat with pain when swallowing.
 - Swollen cervical lymph glands.
 - Headache. , Temperature greater than 100F (37.8 °C)

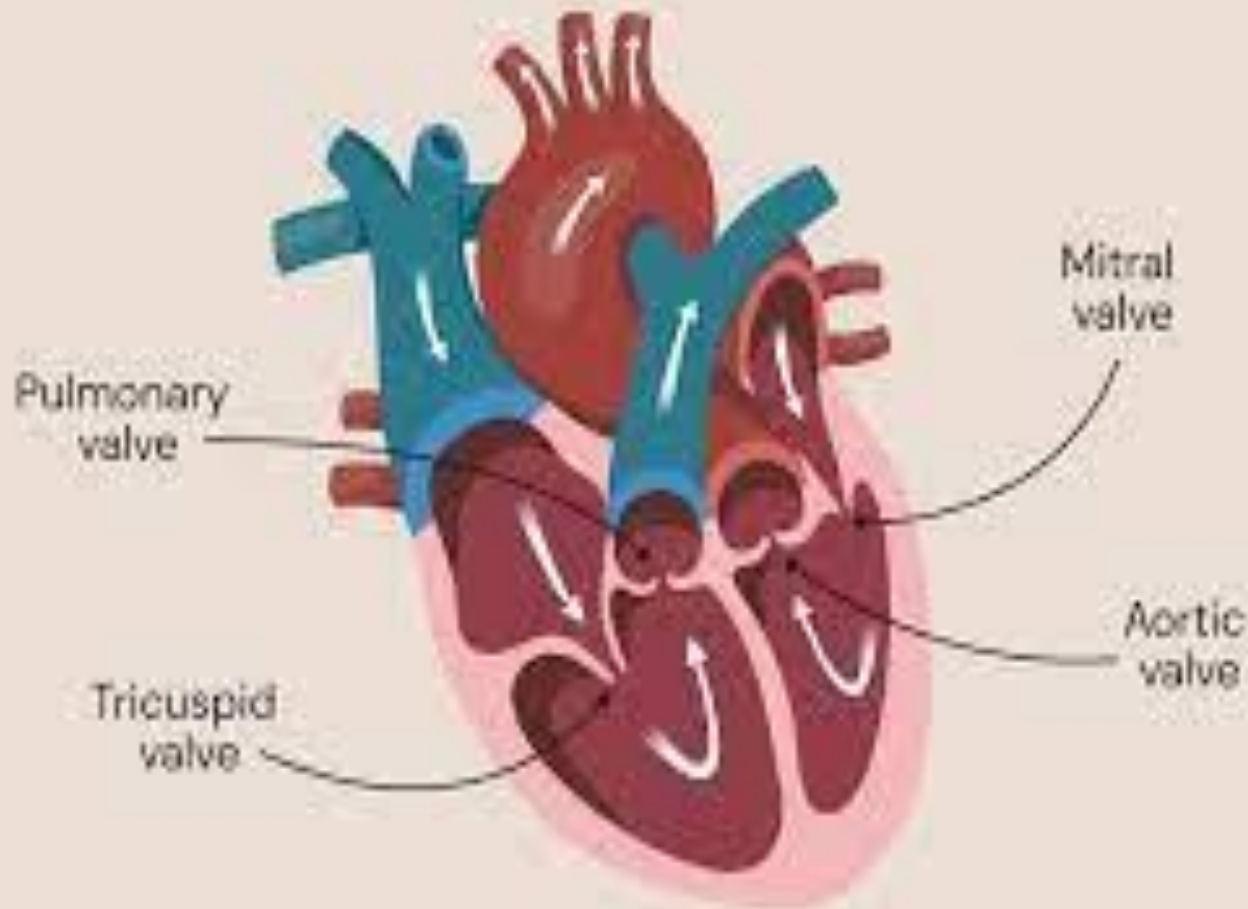
Explain to patient:

- Anticoagulation use

need for frequent lab monitoring.

- Avoid contact with people has a RTI
- Maintain good dental hygiene.
- Check the physician, if detect signs of heart failure:
shortness of breath, weight gain, nonproductive cough.
- Return to normal activities slowly

2- Management of patient with Valvular heart disease



Outline:

- Introduction
- Etiology
- Clinical Features
- Diagnosis
- Complications
- Treatment
- Prevention

Definition

Any **dysfunction** or **abnormality** of **one** or **more** of the heart's **valves** . The valves may become **Stenotic** or **Regurgitant**.

Symptoms depend on the specific **valve** involved and the **severity** of dysfunction



SYMPTOMS OF HEART VALVE DISEASE



Shortness of
Breath



Heart
Palpitations



Swollen Ankles,
Feet or Abdomen



Weakness or
Dizziness



Rapid Weight
Gain



Chest
Discomfort



Fatigue



Fever, Chills, or
Body Aches



Consult Our Experts Today!

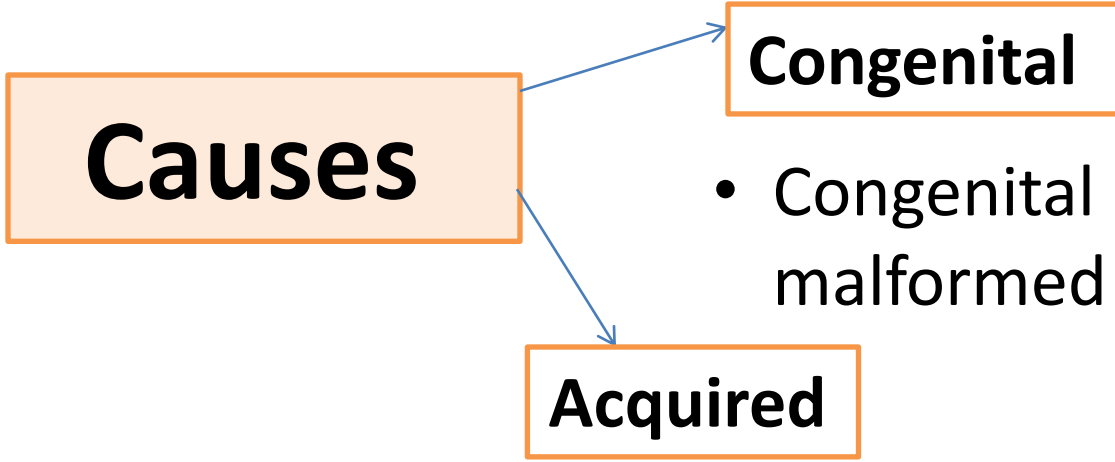
General Symptoms

- **Shortness of breath** especially during exertion or when lying flat (orthopnea).
- **Fatigue** and **weakness**
- **Palpitations**
- **Chest pain** or discomfort
- Dizziness or syncope (**fainting**)
- **Swelling** in ankles or abdomen (**edema**)
- Rapid **weight gain** due to fluid retention

Risk factors

- Age > 65 years
- History of rheumatic fever
- Infective endocarditis
- High blood pressure
- Coronary artery disease
- Congenital heart defects
- Autoimmune disorders
- Radiation to the chest
- Intravenous drug use (risk of infective endocarditis)
- Hyperlipidemia, diabetes, smoking

Causes



```
graph TD; A[Causes] --> B[Congenital]; A --> C[Acquired];
```

Congenital

- Congenital heart defects , malformed valves at birth

Acquired

- **Rheumatic fever** : (especially mitral valve)
- **Age**-related degeneration – calcification.
- **Endocarditis** : bacterial infection damaging the valves
- Ischemic heart disease (**IHD**).
- **Radiation** therapy to the chest
- **Autoimmune** diseases (lupus, rheumatoid arthritis)
- **Cardiomyopathy**.

Diagnosis of Valvular Heart Disease

Physical examination:

- Presence of symptoms.
- Murmurs on auscultation
- Signs of heart failure or fluid overload.

Diagnostic studies :

- Echocardiograph
- Electrocardiogram (ECG)
- Chest X-ray
- Cardiac Catheterization
- Cardiac MRI
- Cardiac CT

Prevention

- Treat **strep throat**
- Good **dental hygiene** (reduces endocarditis risk)
- Prophylactic **antibiotics** before certain procedures (for high-risk individuals)
- **Healthy lifestyle**: exercise, balanced diet, avoid smoking and alcohol
- Control risk factors: manage hypertension, diabetes, and cholesterol
- Regular cardiac checkups (especially if there's a murmur or family history).

Treatment

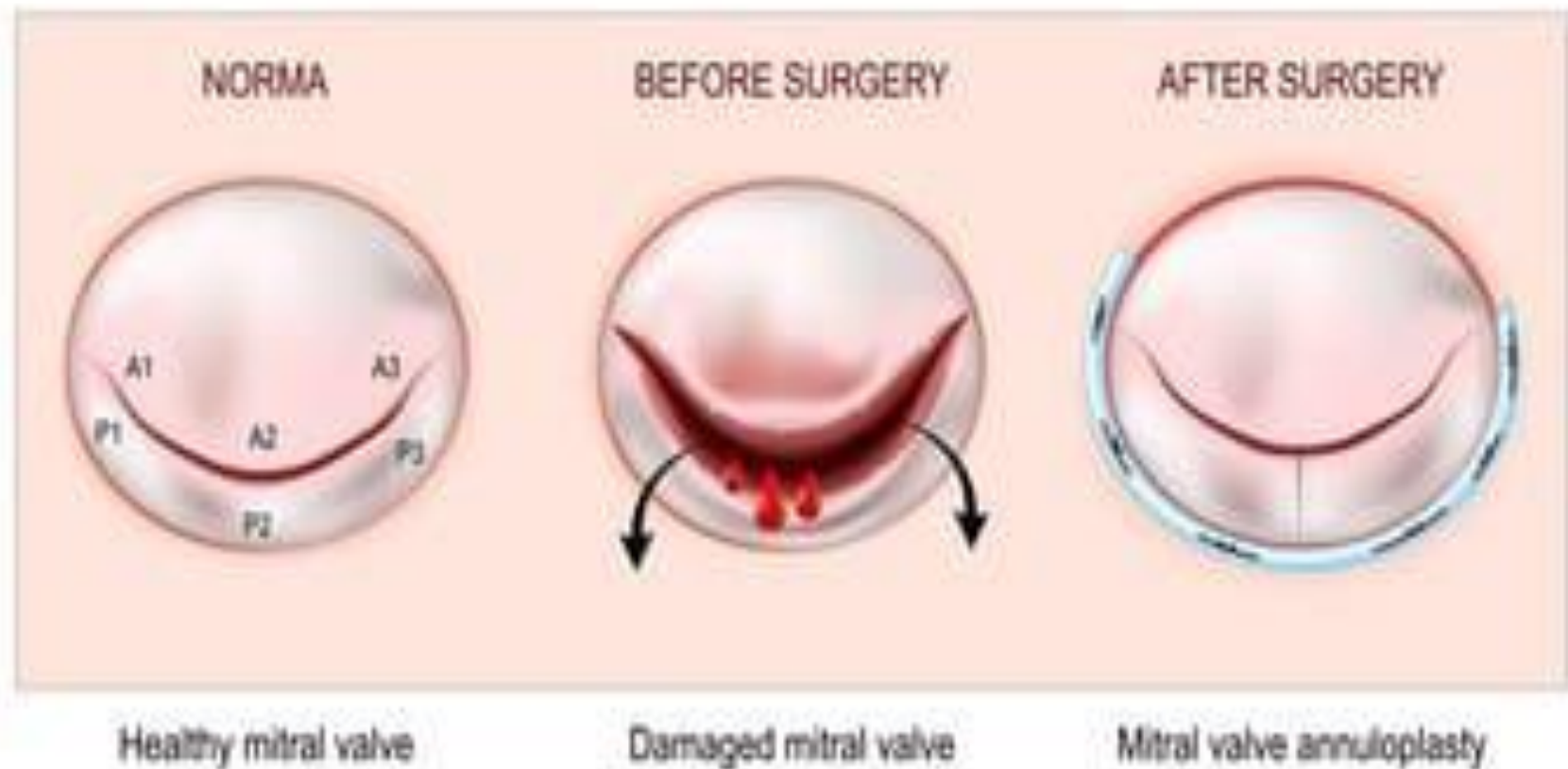
Medical

- Diuretics
- Beta-blockers
- Anticoagulants

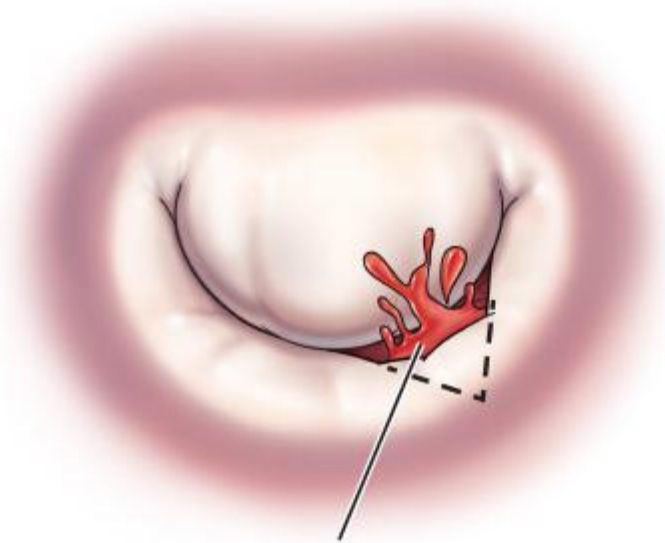
Surgical

- Valve repair
- Balloon valvuloplasty
- Transcatheter Aortic Valve Replacemet

Mitral valve repair surgery

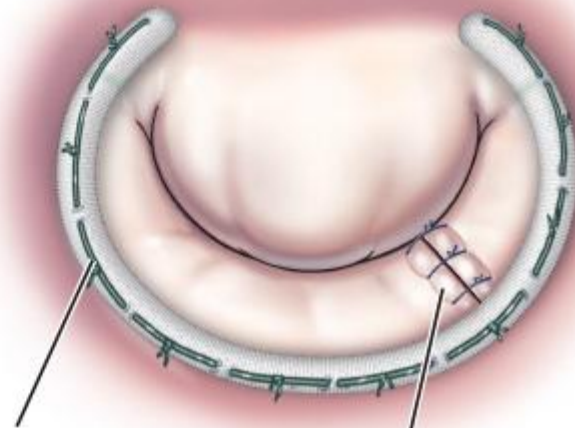


Mitral valve prolapse



Regurgitation

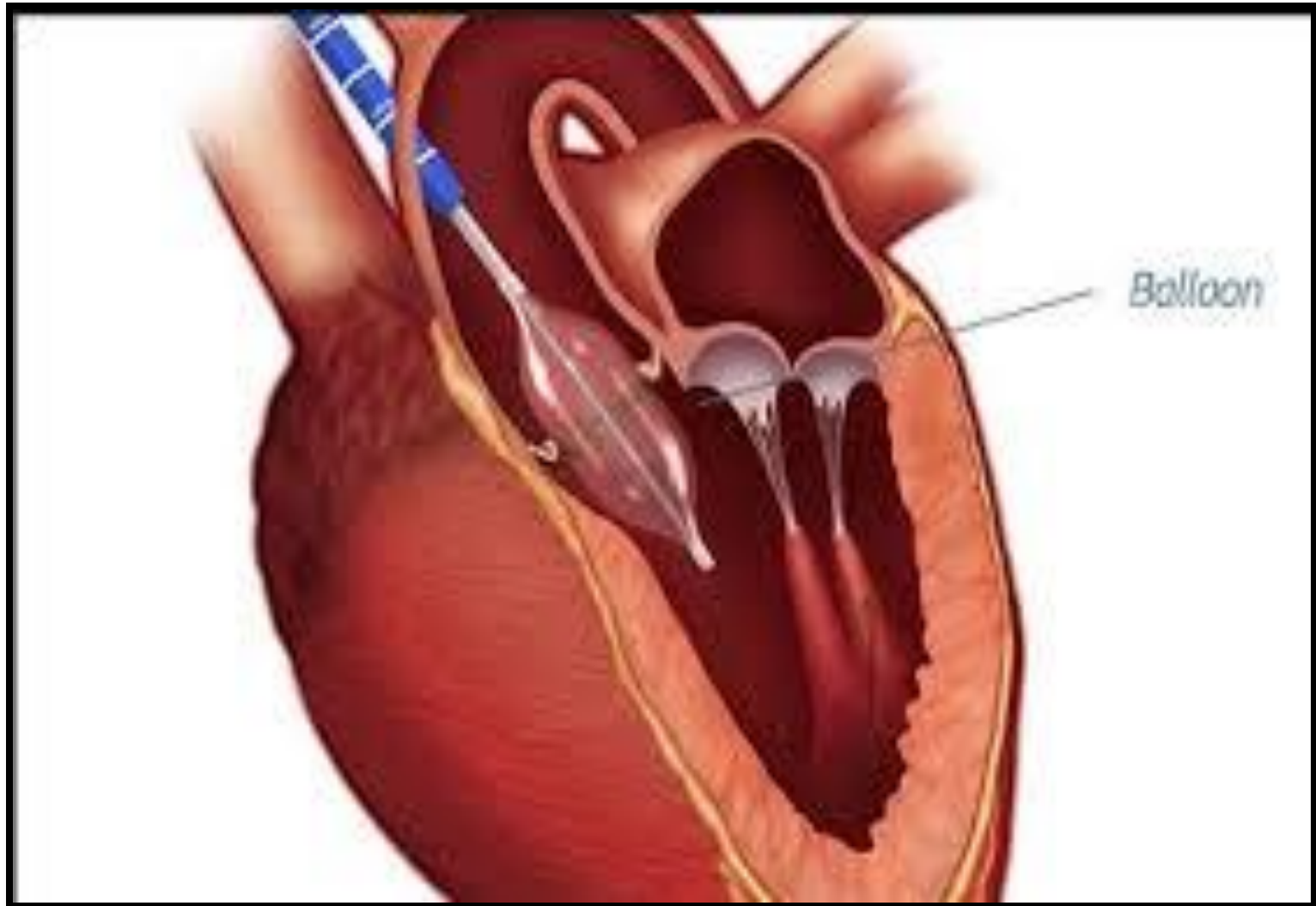
Mitral valve repair



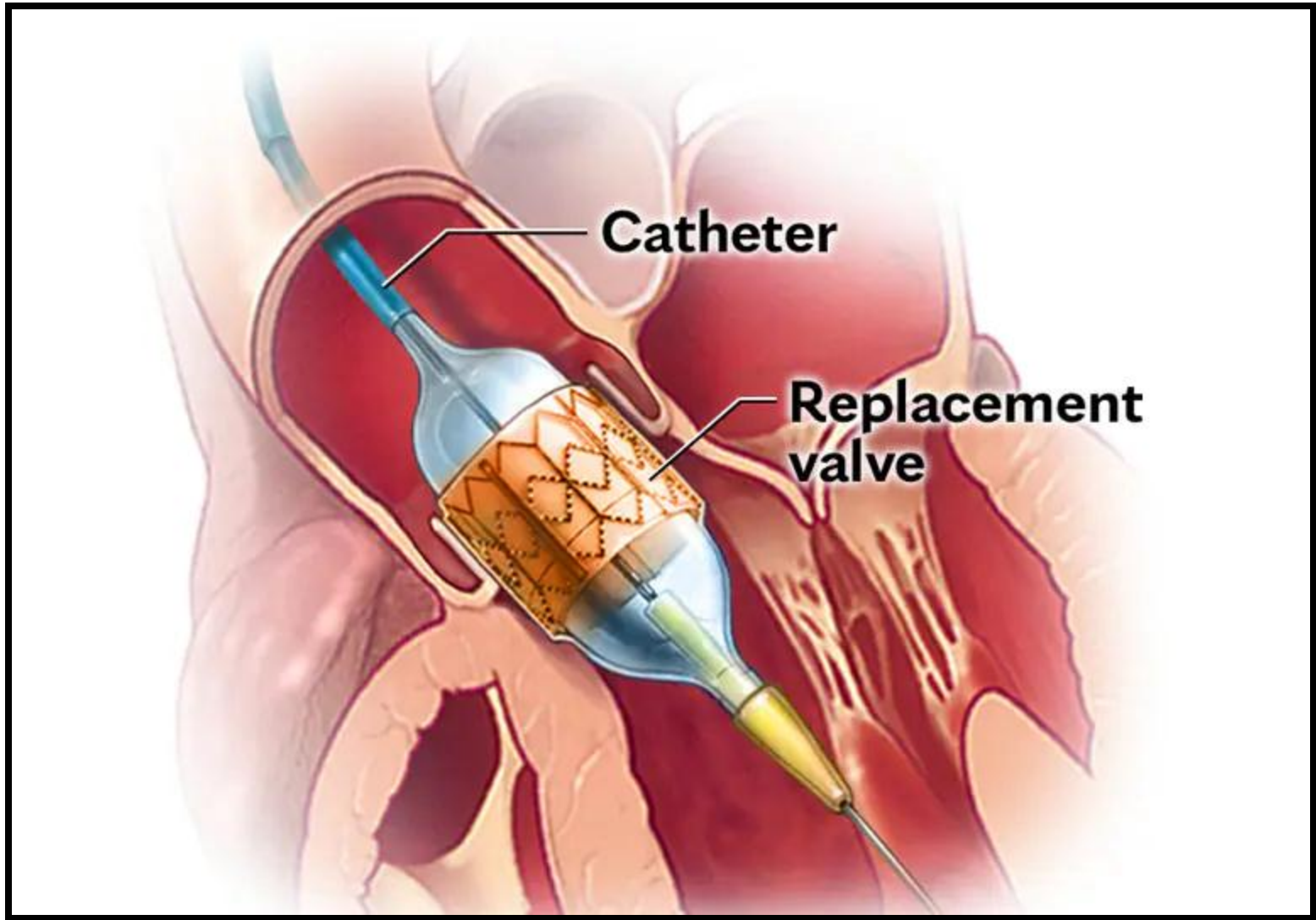
Annuloplasty
ring

Resection

Balloon valvuloplasty



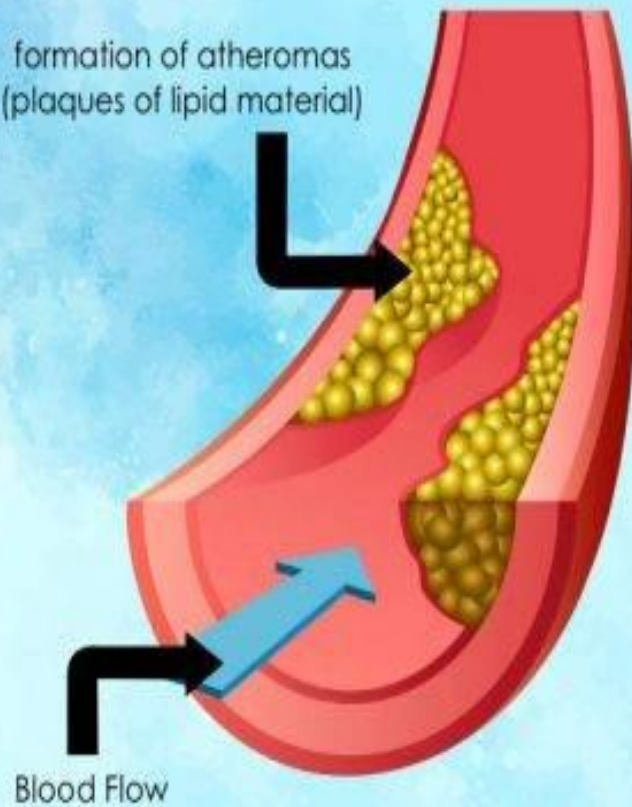
Trans catheter Aortic Valve Replacement



3- Management of patient with Arteriosclerosis and Atherosclerosis

ATHEROSCLEROSIS

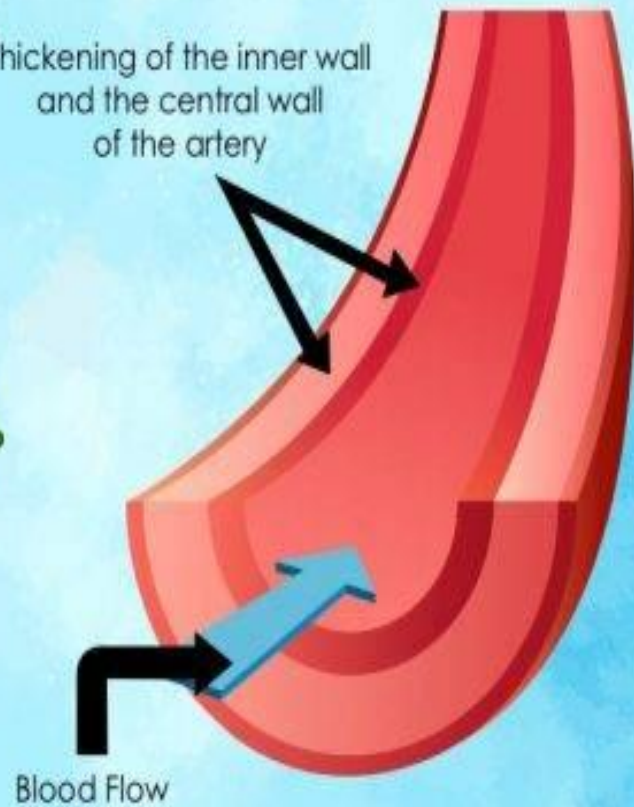
formation of atheromas
(plaques of lipid material)



VS.

ARTERIOSCLEROSIS

thickening of the inner wall
and the central wall
of the artery



- Atherosclerosis is a chronic condition characterized by the buildup of **fatty** deposits (plaques) **inside the walls** of arteries, leading to **thickening, hardening, and loss of elasticity** of the arterial walls.

Normal artery



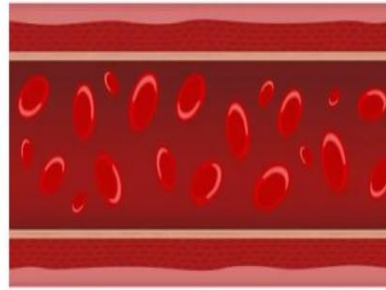
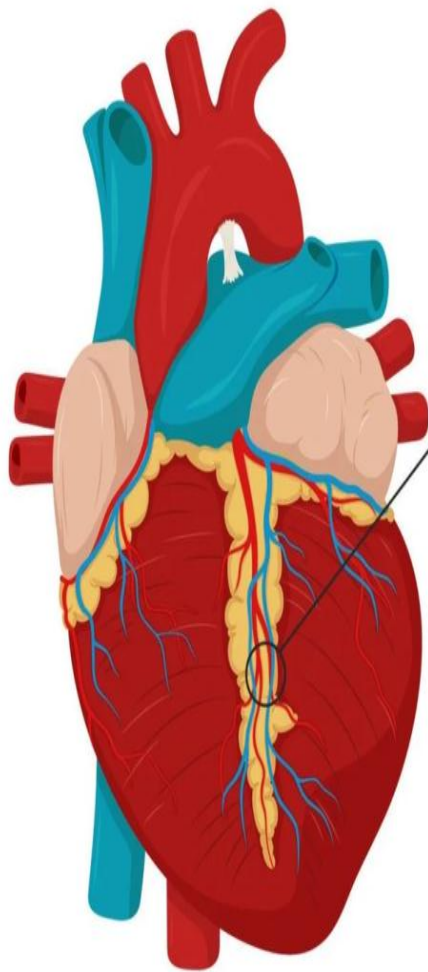
Blood flow

Artery narrowed by atherosclerosis

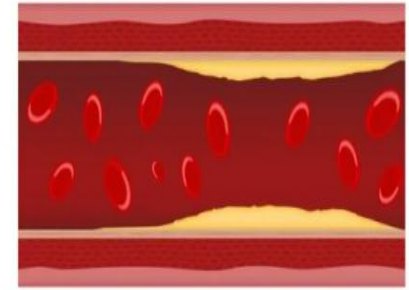
Plaque



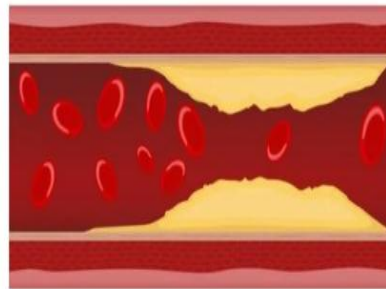
This process narrows the arteries, restricting blood flow and potentially causing cardiovascular diseases such as coronary artery disease and stroke.



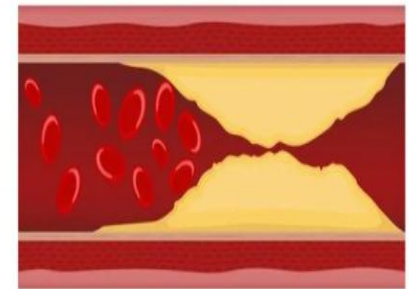
Healthy artery



Beginning of plaque formation



Increased plaque accumulation



Narrowed artery blocked by a blood dot

Causes

Modifiable

**High
cholesterol
levels**

Smoking

Obesity

**Sedentary
lifestyle**

**Unhealthy
diet**

Stress

Causes

Non Modifiable

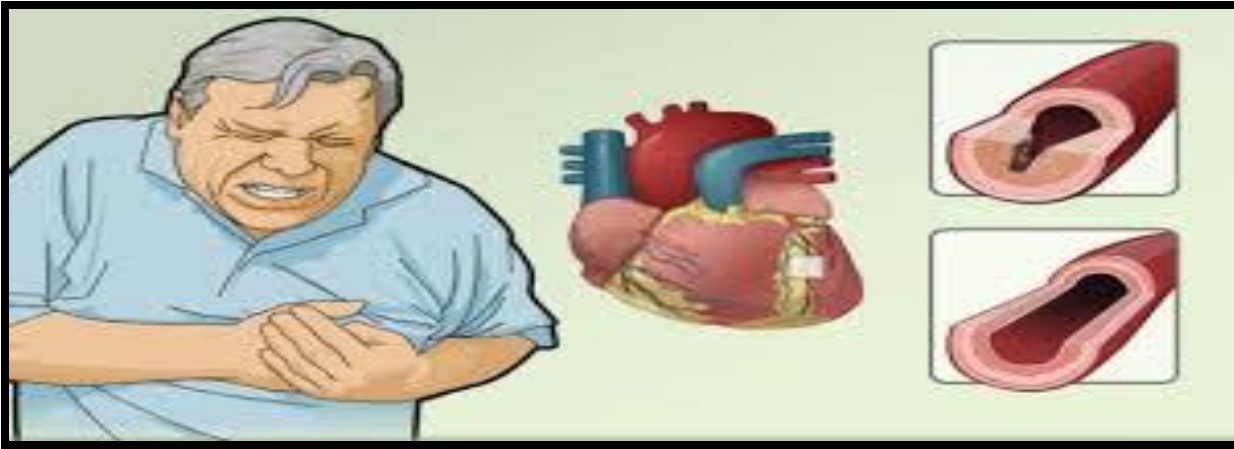
HTN

DM

Genetic

Age

Clinical symptoms



**Chest discomfort
during exercise or
emotional stress.**



**Leg cramps
when
walking.**



**Shortness
of breath with
light activity.**

Prevention

- **Healthy diet**
- **Regular exercise**
- **Quit smoking**
- **Maintain healthy weight**
- **Control blood pressure and regular monitoring for blood glucose level .**
- **Stress management**

Treatment

Medical

- Anti hypertensive
- Antiplatelet agents
- Diabetes management drugs

Surgical

- Angioplasty with stent placement
- Coronary artery bypass grafting (CABG)

Angioplasty



Stent Delivery System In Place

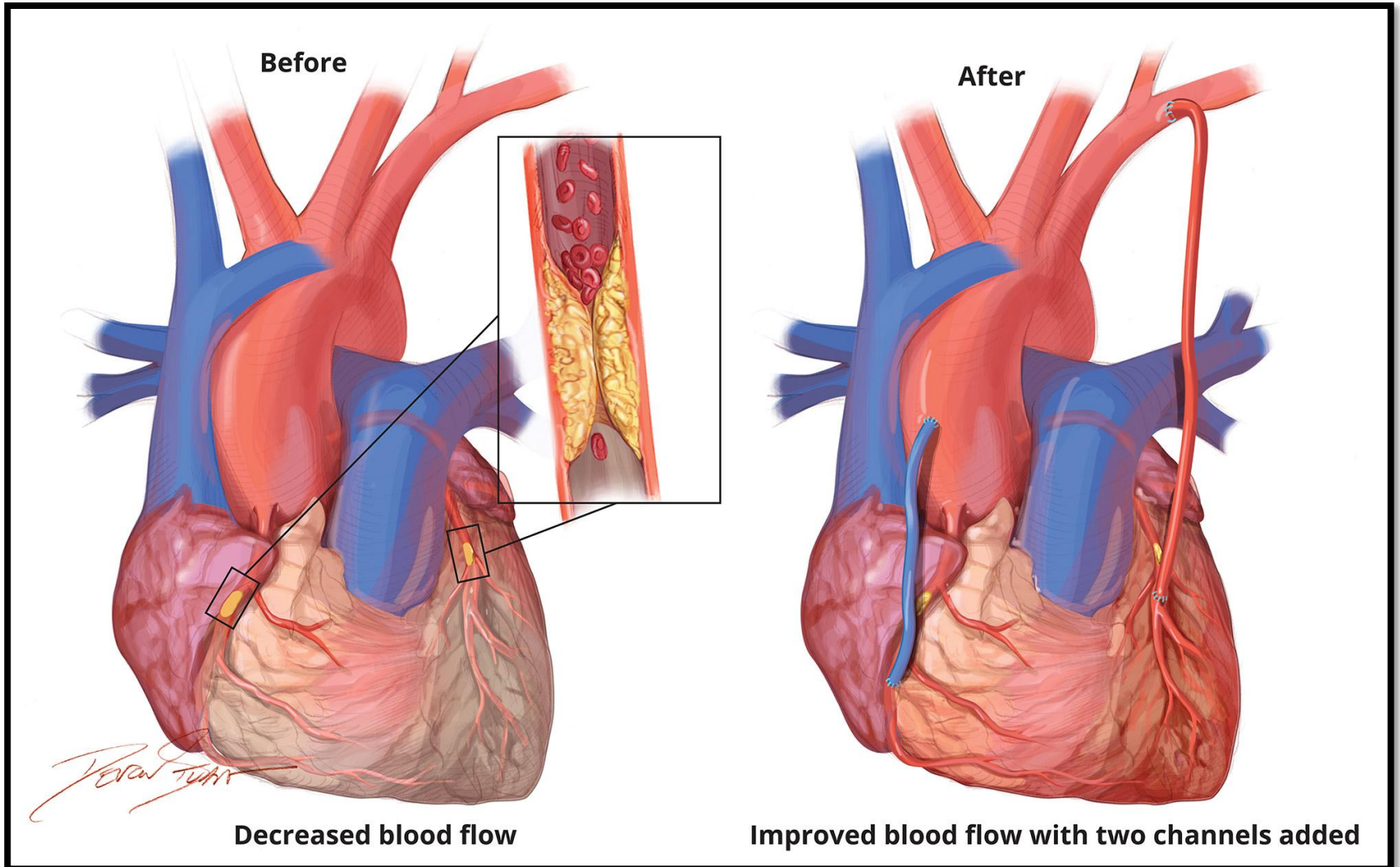


Stent Expands As Balloon Inflates



Catheter Removed, Stent Implanted

CABG



Nursing diagnosis

- Ineffective Tissue Perfusion related to arterial occlusion due to atherosclerosis.
- Activity Intolerance related to decreased o₂ supply to tissues.
- Imbalanced Nutrition related to unhealthy eating habits.
- Risk for Ineffective Health Management related to lack of knowledge about disease and treatment.



Summary



thank
you